

AMRO AL-BAALI

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SKILLS

Domain Knowledge: Sensor fusion, state estimation, SLAM, pose graph optimization, computer vision, matrix Lie groups, linear algebra.

Programming Languages: C++, Python, MATLAB.

EDUCATION

M. Eng Mechanical [McGill University](#)

📅 05/2019 - 08/2021

📍 Montreal, Canada

- CGPA: 3.77/4.00. Supervisor: [Prof. J. R. Forbes](#).
- Thesis title: *Augmenting Sensor Measurements with INS Estimates in a Graph Based SLAM Problem*.
- Publication: A. Al-Baali, T. Hitchcox and J. R. Forbes, "Combining DVL-INS and Laser-Based Loop Closures in a Batch Estimation Framework for Underwater Positioning," in *IEEE Journal of Oceanic Engineering*, doi: [10.1109/JOE.2023.3286854](https://doi.org/10.1109/JOE.2023.3286854).

B. Eng Honours Mechanical,

Minor in Computer Science [McGill University](#)

📅 09/2014 - 04/2019

📍 Montreal, Canada

- CGPA: 3.83/4.00. Supervisor: [Prof. J. R. Forbes](#).
- Thesis title: *Parallel Feedforward Control Using Linear Matrix Inequalities*.

WORK EXPERIENCE

Senior Engineer & Team Lead - Scene Perception [Pickle Robot](#)

📅 02/2024 - Present

📍 Cambridge, MA

- Tech lead the Scene Perception sub-team, leading 4-5 perception engineers to prototype, design, implement, and scale the Scene Perception sub-system. This included designing interviews to hire members, hiring and onboarding new members, and scaling the throughput of the Scene Perception sub-team to meet tight timelines. The Scene Perception sub-team is responsible for multi-modal sensors' extrinsic calibration and scene understanding, leveraging sensor fusion, computer vision, and lidar-based localization and mapping.
- Worked with system architects and autonomy sub-team leads to design the Scene Perception architecture that satisfies the system constraints and delivers the desired performance. Specifically, the Scene Perception sub-system provides the capability for Navigation and Motion Planning to safely drive and pick packages without hitting obstacles that could damage the robot. This included sensor evaluation and selection, software prototyping of the collision avoidance perception, and sensors extrinsic calibration.
- Maintained and improved the Navigation Perception sub-system for deployed robots. Enhanced the performance of the sub-system by introducing software tools that aid in catching bugs, adding features such as lidar odometry, and redesigning the architecture to enhanced the robot's driving performance.
- Worked with Deployment Engineers to ensure Scene Perception related processes such as sensor extrinsic calibration are clear, and to get feedback on the system performance in the field.

Software Engineer - Robotics Team [Vention](#)

📅 05/2023 - 01/2024

📍 Montreal, Canada

- Designed and implemented a new feature that brought a new types of users (power users) that are interested in directly controlling the robot through programming it. I worked on the architecture that allows the user to simulate the commands before executing them on the robot, allowing for faster prototyping for the customer.
- Worked closely with customers to introduce the new feature to them and customize it for their needs. I provided periodic updates including prototypes the customers can test on.
- Improved the process of prototyping on **ROS**, which was used to run the robot simulations, by automating the build and deployment processes. My changes reduced friction in the development process, allowing developers to prototype more quickly and to introduce less bugs.

Software Engineer - Localization and Mapping [Avidbots](#)

📅 09/2021 – 03/2023

📍 Kitchener, Canada

- Contributed to the maintenance and development of lidar-based localization and mapping pipeline, improving the non-linear least squares algorithms used in state estimation and scan-matching.
- Contributed to the development of new features, including localization close to walls (enabling wall following), and visual-based localization for improved initial-localization.
- Led the effort of adapting wheel odometry pipeline to accommodate the removal of wheel encoders to reduce the robot cost. This involved developing a new calibration routine to calibrate the steering wheel encoders and changing the kinematic model.
- Led the effort of setting up an OptiTrack system to evaluate the localization performance during in-house tests. The effort included the design and implementation of a novel calibration routine that aligned the OptiTrack coordinate system with the map used in the localization pipeline.

Graduate Student - SLAM [DECAR group \(McGill University\)](#)

📅 05/2019 – 08/2021

📍 Montreal, Canada

- Collaborated with [Voyis](#) and [Sonardyne](#) to develop a SLAM algorithm for an Autonomous Underwater Vehicle (AUV) equipped using a third-party Inertial Navigation System (INS) and the [Voyis Insight Pro](#) high-precision laser scanner.
- Researched and designed a novel localization protocol that treats the INS as a black box and fuses its output with Voyis' high-precision laser scanner, which was used to generate highly precise maps of the seabed used by Voyis for various applications. The algorithm leveraged convex and on-manifold optimization.

Mechanical Engineering Intern [MY01](#)

📅 05/2018 – 04/2019

📍 Montreal, Canada

- Designed and executed mechanical tests on the MY01 device to obtain medical certification, including FDA certification.
- Wrote the software that communicates and controls the testing platform, including moving the test head for penetration testing, generating plots, and writing a GUI for the user to control the device.
- Customized the Autodesk Vault software tool used for CAD version control, customizing it to automatically generate reports and maintain version numbers for released designs.

Undergraduate Research Assistant [McGill University](#)

📅 05/2017 – 08/2017

📍 Montreal, Canada

- Researched the use of multi-objective optimization methods, such as genetic algorithms, for design-space exploration (DSE), under the supervision of [Prof. D. Varro](#).
- Used [OpenMDAO](#) and [Platypus](#) packages to implement the optimization for the DSE.

Teaching Assistant [McGill University](#)

📅 09/2017 – 04/2021

📍 Montreal, Canada

I was a teaching assistant for the following courses, providing tutorials, grading assignments, mid-terms and quizzes, and assisting to students to understand the course materials and assignments.

- [MECH 513 \(Control Systems\)](#) (Winter 2021)
- [MECH 309 \(Numerical Methods\)](#) (Fall 2019)
- [MECH 412 \(System Dynamics and Control\)](#) (Fall 2017)